

Input vs. Output

The model takes as input unstructured, noisy text descriptions from real-world financial consumer complaints, with the narratives stripped of non-text records. It predicts an operational issue classification, assigned to one of 43 target classes, paired with a calibrated confidence score for automated triage.

Classification Problem

This is a single-label, multi-class text classification task. The 43 classes are mapped directly from the dataset's Issue field and encoded with scikit-learn's LabelEncoder. They represent real-world operational silos in financial services (e.g., billing, fraud, card management), which makes routing more efficient.

Labeling

Labels are derived from the structured Issue field provided by consumers and financial institutions in public repositories. Consistency and accuracy are checked by filtering out incomplete entries that lack text narratives, encoding targets systematically, and evaluating performance with stratified splits, Macro F1-scores, and confusion matrices to detect class overlap.

Data Creation

The training and evaluation corpus comes directly from the public Consumer Financial Protection Bureau (CFPB) Consumer Complaint Database (complaints.csv.zip). A curated, independent African-English complaint corpus is planned for future cross-domain generalization testing. All data consists of direct, real-world consumer complaints. Synthetic generation is not used, though class weighting or oversampling may be applied to handle class imbalance.

Web Collection Ethics

Data is sourced from public open-access repositories, primarily the CFPB Consumer Complaint Database, using custom Python scripts for data cleaning, filtering, and LabelEncoder target processing. Consent, copyright, and privacy have been addressed: the dataset consists of publicly released records in which personally identifiable information (PII) has been scrubbed, so PII is not intentionally used as a training feature.

Bias and Representation

The dataset may leave out consumers who submit complaints in languages other than English, non-written complaints (such as call transcripts without text), and regional dialects or phrasing that are underrepresented in standard US datasets. To make the dataset more representative, models will be validated on real-world, noisy consumer language and tested against an external African-English complaint corpus to assess cross-domain reliability across regional phrasing.

Values and Community Alignment

The dataset design prioritizes fairness, accountability, and consumer protection by focusing on real user grievances and by evaluating rare, high-severity issues (like fraud) with balanced metrics rather than raw accuracy. Consumers benefit from faster, more accurate dispute triage. Accountability is maintained through a human-in-the-loop fallback: low-confidence predictions and high-risk cases are automatically escalated to human representatives rather than misrouted.